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are four primary (basic) tissue types that interweave to form the 'fabric' of the body. They are, the epithelial tissue (covering), the connective tissue (support), the muscle tissue (movement) and the nervous tissue (control) (Figure 3.1).

3.2 Epithelial Tissue

Epithelial tissue is a sheet of cells that covers the body surface or lines the body cavity. It occurs in the body as a **covering, as a lining epithelium and as glandular, epithelium**. The functions of epithelium includes **protection, absorption, filtration, excretion, secretion** and **sensory reception.** Based on the structural modification of the cells, the epithelial tissues are classified into simple epithelium and compound epithelium or stratified epithelium.

Simple epithelium is composed of a single layer of cells. They are found in the organs of absorption, secretion and filtration. Simple epithelial tissue is further classified into squamous epithelium, cuboidal epithelium, columnar epithelium, ciliated epithelium and pseudostratified epithelium (Figure 3.2). The **squamous epithelium** is made of a single thin layer of flattened cells with irregular boundaries. They are found in the kidney glomeruli, air sacs of lungs, lining of heart, blood vessels and lymphatic vessels and are involved in functions like forming a diffusion boundary and filtration in sites where protection is not important. The **cuboidal epithelium** is made of a single layer of cube like cells. This tissue is commonly found in the kidney tubules, ducts and secretory portions of small glands

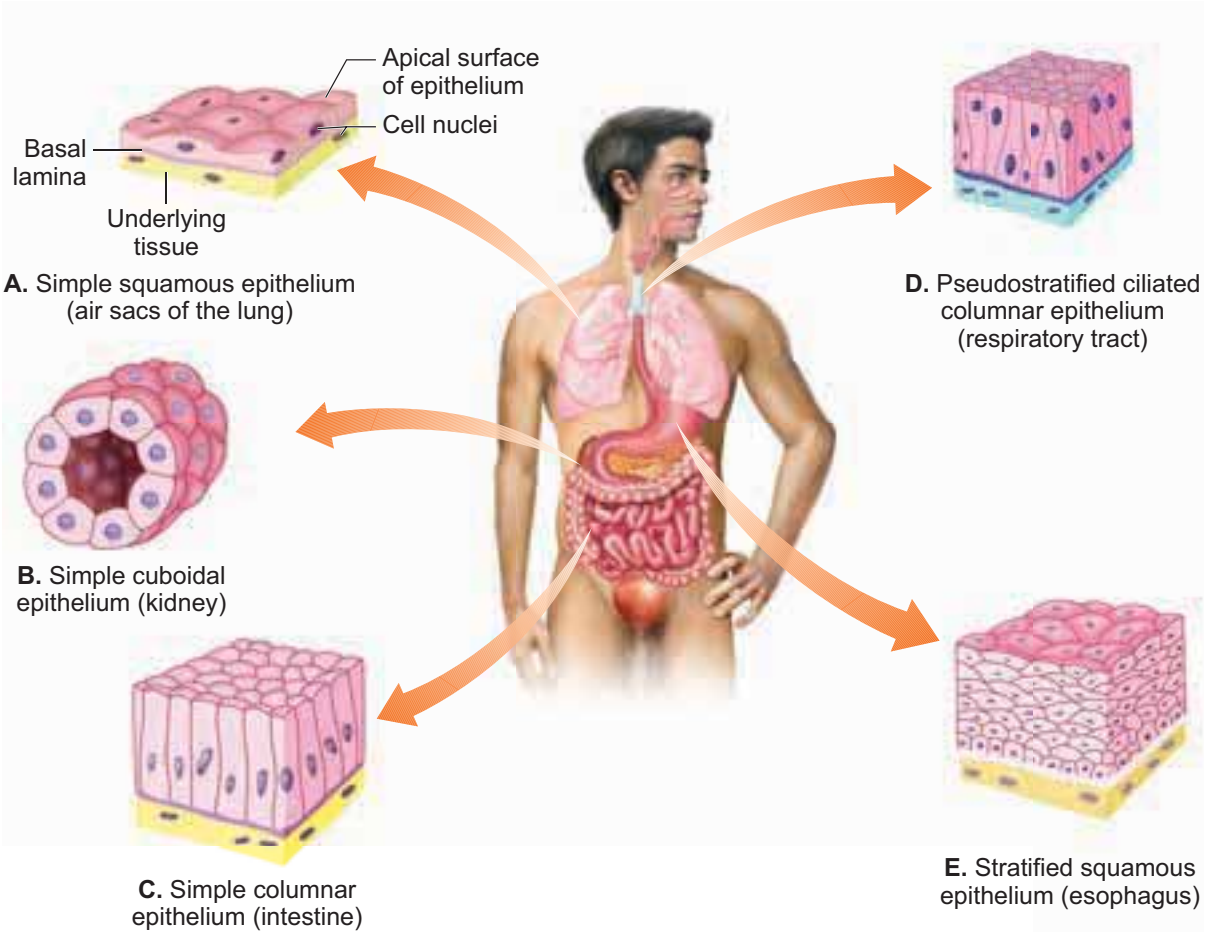


Figure 3.2 Types of Epithelial tissues